

THINK RIGHT

What we're talking about is the tinkering mindset, and applying that to your life in general. We don't want you to take apart an important machine at the office to try and figure out how it works... that will probably get you fired, or even jailed, depending on how important and expensive the machine is... Instead, what we want you to do is to learn to take apart your life, or your job, or today's task, and study it as a tinkerer would. Because we know our own lives, we have valuable insight into how it can be broken down.

A simple example is the life of a technology writer, who has to juggle many balls in the air at once. There's planning of future articles, loads of reading to be done, research, speaking to people, meetings to attend, reviewing gadgets, discussions with peers, actually writing stuff, gaming, making videos, being on top of the latest news, even more reading, even more research, gaming, tinkering with our PCs, engaging on social media, watching cat videos, gaming, attending press conferences, interviewing people, going for junkets like CES Las Vegas, going out into the market and speaking to vendors, keeping an eye on sales and offers, gaming, voicing over videos, trying to find time to read a book or three, scouring the app stores for interesting stuff, trying to stay healthy and fit enough to survive a decade or two, listening to family members complain about you not spending time with them, and if we get any free time after all of that... gaming.

There are a myriad of ways in which we can tinker and improve on how we do things, or even better, automate certain tasks. And this is the point in such discussions where it becomes mandatory to talk about AI.

JUDGEMENT DAY

All of us know that a general AI is coming, and we also know that it will cause jobs to change drastically. There are aspects of all of our jobs that AI, or a machine of some form, is just better suited to do. Anything repetitive or anything that can be templated is just begging for machines to do it. And of course, machines will do it.

Machines should be doing it. You can get machines to help you do all of that right now anyway, which will free up your time and allow you to do more of the stuff that machines cannot do.

As humans, our true worth is not to do what dumb machines already can, but to outthink machines, to do things better, to innovate, and of course, to tinker. Most jobs today already need us to use our brains, and if you're reading Digit, it means you're probably using your brains far more than the average Joe does. In such jobs, working with software is usually the norm, but it's not like we can hack (tinker with) that software to work better for us. Instead, what we can do is look at everything else we work with in the office and try and break those down into simpler tasks, and then find ways in which AI can help us.

Of course, you need to know how to dabble in some code to achieve anything with AI, but thankfully, it's getting a lot easier to do thanks to software packages and services that are available to us.

HACK YOUR BRAIN

And now we get to the part where you learn how to hack your life. You do this by engaging in what's called computational thinking (CT), which is a way to break down tasks and problems into smaller chunks, translate that into something a computer could understand, and then use a computer to help you solve aforementioned problems. In essence, it is what enables computers to help us do tasks that are usually specific to our jobs. Remember the introduction to this article? Where we talked about learning a new language? Think of CT as learning a new language, where you learn to talk and think like a computer would.

And it's not just computers that can be helped by using CT, but us as well. When you break down a problem, or organise data logically, and try and convert it into simple yes or no answers (binary) often, it helps you arrive at a solution all on your own.

CT can be achieved by breaking any problem you have into three main chunks. The first is called Abstraction, which is essentially you decom-

posing a problem into many different chunks that allow you to formulate the problem in computer speak. Then comes Automation, which is a way to express how you would arrive at a solution, again in the computer language of your choice – it's like building a model by using data sets. Then comes Analyses (plural) which is the writing and testing of algorithms to arrive at a solution. To simplify and dejargonise the idea, let's look at a simple math example:

Let's say you have to find the sum of all numbers between 1 and 10,000 (don't Google it!). And of course, you can't just use a computer and find out, because you're trying to think of this problem like a computer.

If you were to manually add $1+2+3+4+5+6+7...+10,000$, it would take forever! Although a computer could do it this way in a heartbeat, what if we wanted to find the sum of all numbers between 1 and 6,516,516,546,516,546,519,654,872,346,987,287,364,598,712,634,578,926,347,856,243,578? A modern computer could still tell you the answer in a heartbeat, but not because it would do this manually, by adding all those numbers together... It's been programmed instead to recognise patterns.

What you'd need to do is to break the problem down into a simpler solution. Instead of using $1+2+3+4+5+6+7...+10,000$, you need to find a better way to do this. Since we have to add all numbers between 1 and 10,000, a simpler way to write this down would be:

$1 + 10,000$
 $2 + 9,999$
 $3 + 9,998$
 Etc.

Here you can see that the answer for all of them is 10,001. Next, you need to know how many of these pairs of numbers you will have before you exhaust the numbers that need to be added. Since you're taking 2 numbers per equation, and there are 10,000 numbers, the answer is $10,000/2$ or 5,000 pairs. Thus, 5,000 pairs of numbers that total 10,001 in a formula is:

$10,001 \times 5,000$, or $(10,000 + 1) \times (10,000/2)$ or $(n/2) \times (n+1) = n^2/2 + n/2$.

